Nicholas Lasagna

Berkeley, CA | (510) 827-7009 | nicholaslasagna@gmail.com | linkedin.com/in/nicholas-lasagna-798118277 | https://github.com/nicholaslasagna

EDUCATION

Texas Tech University Lubbock, TX
Bachelor of Science in Computer Science, Minor: Mathematics | GPA: 3.565 / 4.00 Expected May 2027

Accelerated Program: Enrolled in an accelerated BS/MS Computer Science program, completing graduate-level coursework during an undergraduate degree.

Relevant Coursework: Computer Architecture, Theory of Automata, Engineering Statistics, Software Engineering, Differential Equations, Data Structures & Algorithms, Analysis of Algorithms, Systems & Assembly.

TECHNICAL SKILLS

Languages: Python, Java, Rust, C++, C, C#, x64 Assembly, JavaScript, TypeScript, SQL, HTML/CSS
Infrastructure & Cloud: Linux (Ubuntu), Oracle Cloud Infrastructure (OCI), Docker, Git, REST APIs, CI/CD

Systems & Backend: Distributed Systems, Runtime Debugging, JVM Tuning, System Monitoring, PostgreSQL, Supabase, LLMs

Engines & Frameworks: Unreal Engine 4/5, Unity, Clickteam Fusion 2.5, Next.js, Blender

PROFESSIONAL EXPERIENCE

Imagicast Studios | Berkeley, CA | Co-Founder & Lead Developer | 2021 – Present

- Co-founded and led the development of "Abandoned Horror," an asymmetrical multiplayer game currently in development, owning core systems architecture from initial design through ongoing multiplayer development.
- Designed and implemented distinctive multiplayer roles within a 6v1v1 gameplay loop.
- Engineered engine-level gameplay systems, state management, input handling, and runtime multiplayer logic utilizing Unreal Engine 4/5, C++, and C#.
- Coordinated distributed development workflows using Git and iterative engineering practices, balancing technical correctness, maintainability, and feature delivery across agile, evolving requirements.

SYSTEMS & ENGINEERING PROJECTS

RealFiction Distributed Multiplayer Server Infrastructure (Java, Oracle Cloud, Ubuntu, Distributed Systems)

- Architected and operated a distributed multiplayer server infrastructure hosted on Oracle Cloud Infrastructure using Ubuntu Linux.
- Managed live production backend systems supporting concurrent real-time multiplayer workloads, monitoring availability and latency under real player load.
- Configured advanced networking, permissions, deployment pipelines, process monitoring, JVM tuning, and runtime optimization.
- Maintained multi-server architecture integrating custom backend services, persistent player data systems, and live production debugging.

RealLang (Compiler Engineering, Structured Diagnostics)

- Designed and implemented a working programming language with structured, machine-applicable diagnostics targeted at LLM code generation and repair.
- Built the full pipeline: lexer, parser, type checking, and diagnostic emission, with error messages designed to be consumed by both humans and AI tooling.

RealForge Workbench (Local-First AI Tooling, TypeScript)

- Building a local-first desktop AI engineering workbench that connects to a user-configured, OpenAI-compatible local model provider and keeps AI output inspectable and untrusted until validated.
- Engineered the safety architecture: bounded context, untrusted-by-default output labeling, safe dry-run previews, approval gates, and automated privacy scans before any change is applied.
- Bootstrapped the workbench's starting foundation on the RealLang programming language.

HeartHaven (Next.js, TypeScript, Supabase, PostgreSQL)

- Developing a real-time multiplayer interactive web platform featuring persistent user systems, social interaction, and browser-based gameplay mechanics.
- Building full-stack infrastructure using modern cloud-backed web technologies, implementing authentication, persistent player state, and real-time synchronization.
- Designing scalable database architecture with Row-Level Security (RLS) for progression systems, inventories, and live user experiences.

Runtime Tooling & Low-Level Systems Project (Rust, Memory Safety, Concurrency)

- Engineered performance-sensitive runtime tooling in Rust for a game modification project with an active community.
- Debugged complex runtime behavior involving concurrent execution and systems-level resource management.
- Focused strictly on memory safety, runtime correctness, and low-level systems interaction to ensure stability under load.

LEADERSHIP & INVOLVEMENT

National Society of Leadership and Success: Active Member.